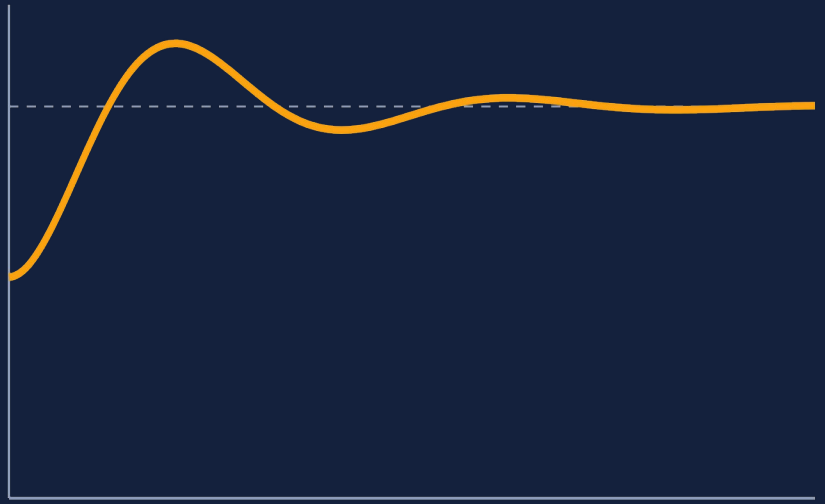
Test Signals and First-Order Response



Control Systems (22EEG53)

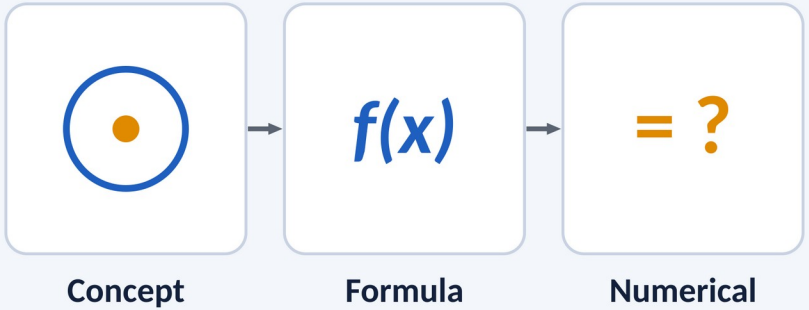
Unit 2: Time Domain Analysis of Control Systems

Topic: Standard test signals; response of zero-order and first-order systems

Prof. Mitesh Astik

Learning Outcomes

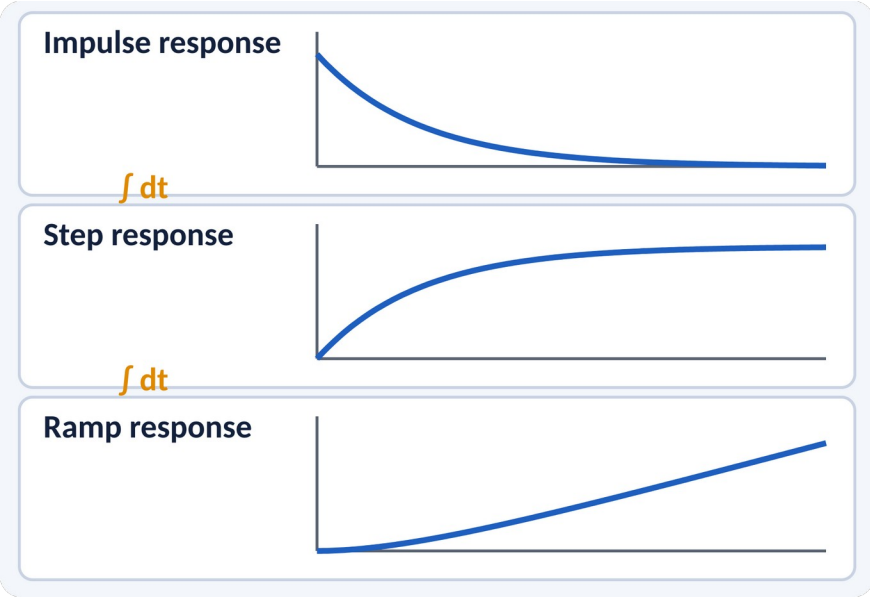
- Name the standard test signals and their Laplace transforms.
- Separate transient and steady-state response.
- Find the response of zero- and first-order systems.
- Use the time constant τ to get specifications.



Linked to CO2
Analyse response of systems to standard test signals.

Recap: What You Already Know

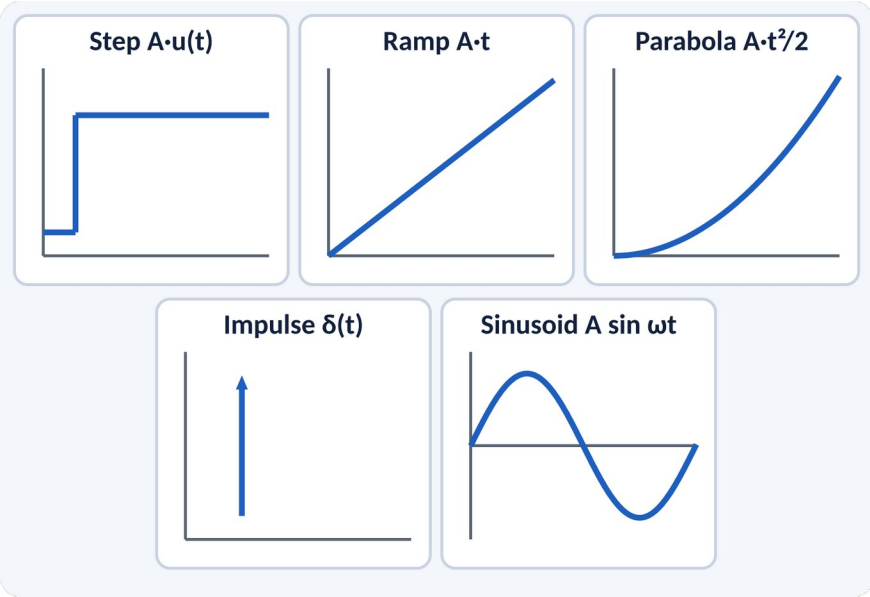
- Laplace transform of unit step is 1/s.
- Inverse transform by partial fractions.
- Poles decide the form of the response.
- Transfer function G(s) = C(s)/R(s).



$C(s) = G(s)\cdot R(s); \quad c(t) = L^{-1}\{C(s)\}$
$L\{e^{(-at)}\} = 1/(s + a)$

Motivation: Why Standard Test Signals?

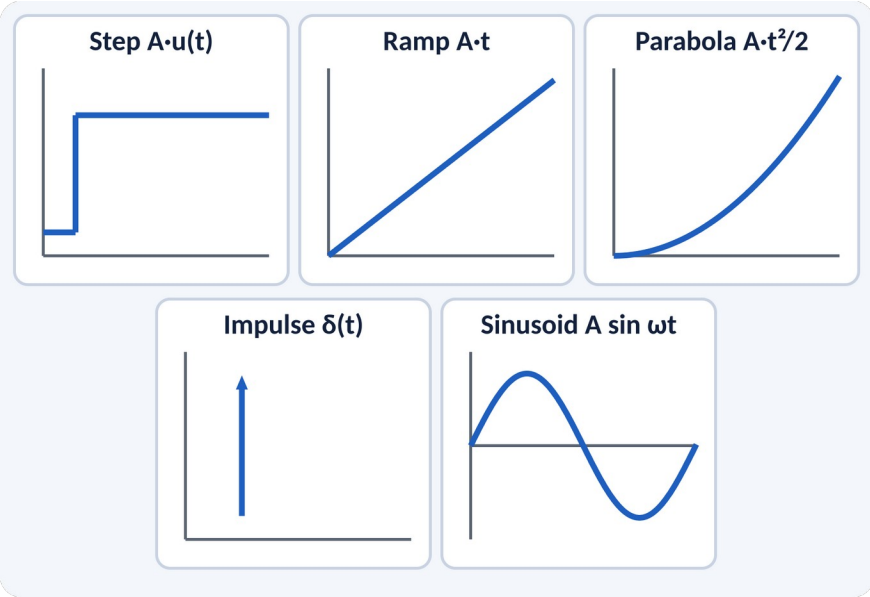
- Real inputs are random and unknown.
- Test signals give a fair comparison of systems.
- Step tests sudden changes; ramp tests tracking.
- Response to one signal predicts the others.



Compare systems with the same input: step, ramp, parabola, impulse, sinusoid

Standard Test Signals

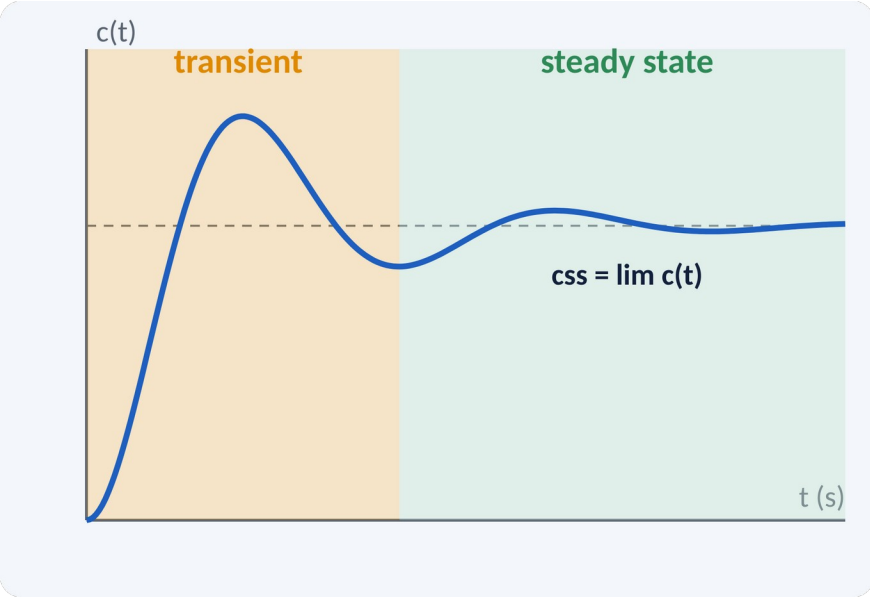
- Step: sudden change of constant size.
- Ramp: constant velocity input.
- Parabola: constant acceleration input.
- Impulse: very short, unit-area pulse.
- Sinusoid: used for frequency response.



$r = A\,u(t) \rightarrow A/s;\ \ A\,t \rightarrow A/s^2;\ \ A\,t^2/2 \rightarrow A/s^3;\ \ \delta(t) \rightarrow 1$
A amplitude; unit signals use A = 1

Transient and Steady-State Response

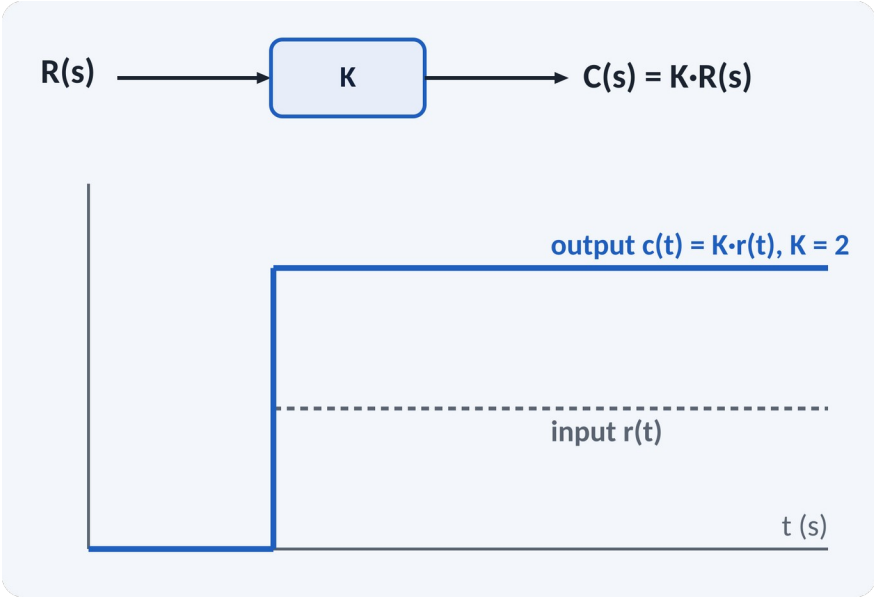
- Total response = transient + steady state.
- Transient dies out with time.
- Steady state is the final behaviour.
- Specifications describe both parts.



c(t) = ctr(t) + css(t), ctr → 0 as t → ∞
css = lim c(t) as t → ∞ (stable systems)

Zero-Order System

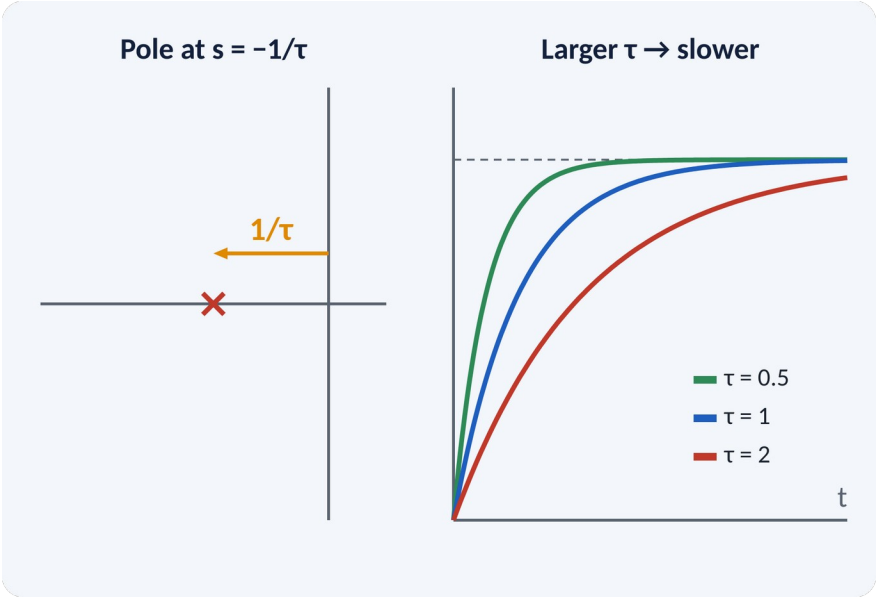
- Transfer function is a pure gain K.
- Output follows input instantly, scaled by K.
- No energy storage, no dynamics.
- Example: potentiometer, ideal amplifier.



G(s) = K → c(t) = K·r(t)
K gain (output unit per input unit)

First-Order System

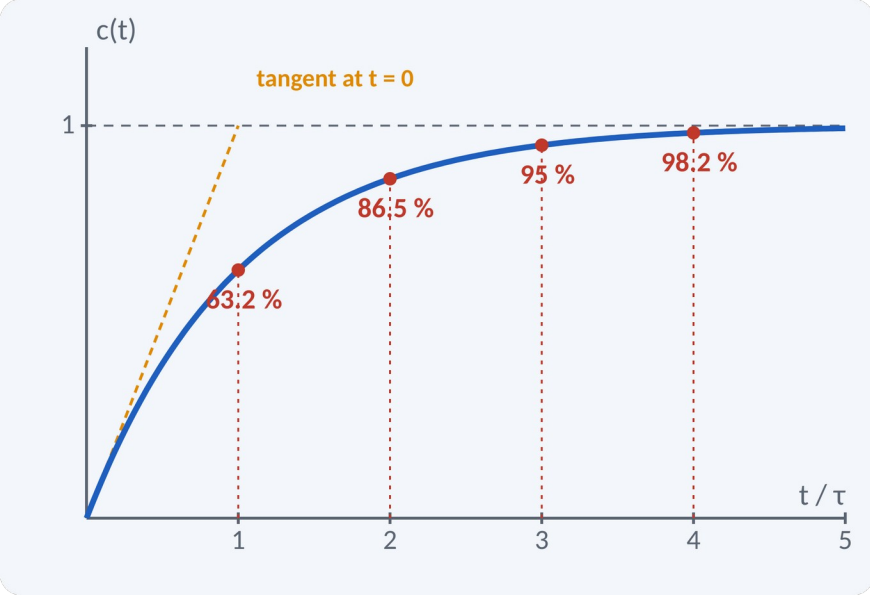
- Contains one energy-storage element.
- Standard form has a time constant τ.
- Single pole on the negative real axis.
- Examples: RC circuit, thermometer, tank filling.



G(s) = K/(τ s + 1); pole at s = −1/τ
τ time constant (s), K DC gain

Unit Step Response of First Order

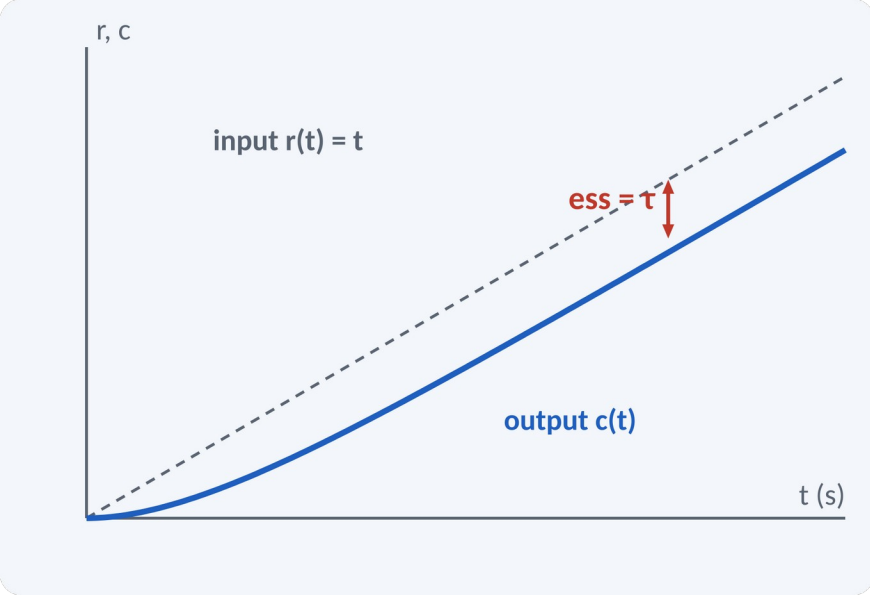
- C(s) = K/[s(τs + 1)]; take K = 1.
- Output rises exponentially to the final value.
- At t = τ it reaches 63.2 % of final value.
- Initial slope is 1/τ.
- Settling: 4τ for 2 %, 3τ for 5 %.



c(t) = 1 − e^(−t/τ); ts(2 %) = 4τ, ts(5 %) = 3τ, tr(10–90 %) = 2.2τ
td = 0.69 τ (time to reach 50 %)

Unit Ramp Response of First Order

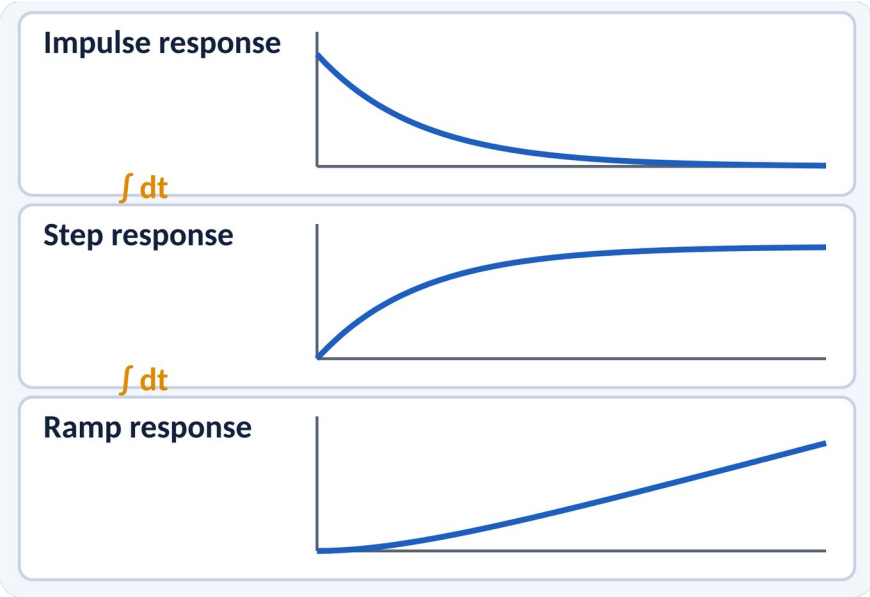
- Input r(t) = t, so R(s) = $1/s^2$.
- Output follows the ramp with a lag.
- Steady-state error equals τ.
- Smaller τ gives smaller tracking error.



c(t) = t − τ + τ e^(−t/τ); ess = τ
ess steady-state error (same unit as the input)

Impulse Response and Relationships

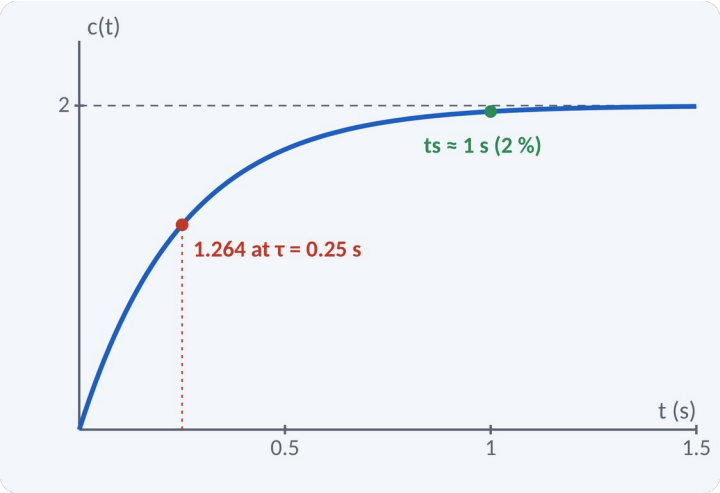
- Impulse: R(s) = 1, so C(s) = 1/(τs + 1).
- Output decays exponentially from 1/τ.
- Step response is the integral of impulse response.
- Ramp response is the integral of step response.



c(t) = (1/τ) e^(−t/τ) (impulse); d/dt(ramp response) = step response
For an LTI system: derivative of input gives derivative of output

Solved Example 1: First-Order Step

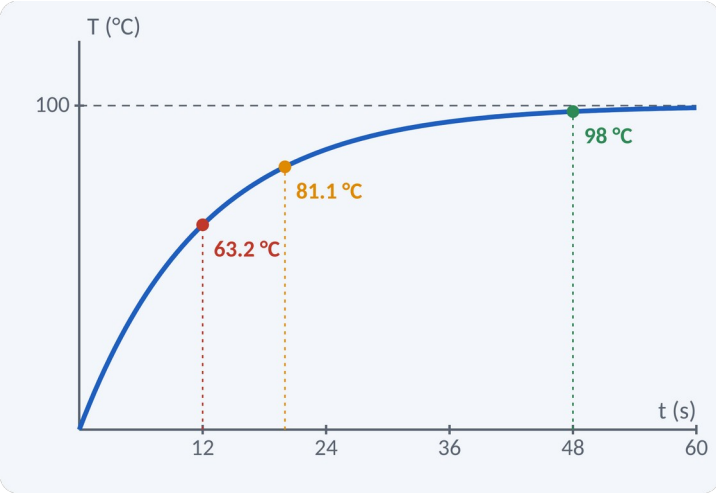
- Given C/R = 4/(0.5s + 2). Find K, τ, c(t) for a unit step, tr and ts.
- 1. Standard form: 4/(0.5s + 2) = 2/(0.25s + 1).
- 2. K = 2, τ = 0.25 s.
- 3. c(t) = 2(1 − e^(−4t)).
- 4. At t = τ: c = 2 × 0.632 = 1.264.
- 5. tr = 2.2τ = 0.55 s.
- 6. ts(2 %) = 4τ = 1.0 s; ts(5 %) = 0.75 s.



Answer: K = 2, τ = 0.25 s, ts = 1.0 s (2 %), c(∞) = 2

Solved Example 2: Thermometer Test

- A thermometer is dipped in 100 °C water. It reads 63.2 °C after 12 s. Find τ, G(s), reading at 20 s, and ts.
- 1. 63.2 % of final value occurs at t = τ, so τ = 12 s.
- 2. K = 1 (reads 100 °C finally): G(s) = 1/(12s + 1).
- 3. T(20) = 100(1 − e^(−20/12)) = 100(1 − 0.189) = 81.1 °C.
- 4. ts(2 %) = 4τ = 48 s.



Answer: τ = 12 s, G(s) = 1/(12s + 1), T(20 s) = 81.1 °C, ts = 48 s

Common Mistakes and Exam Tips

Common mistakes

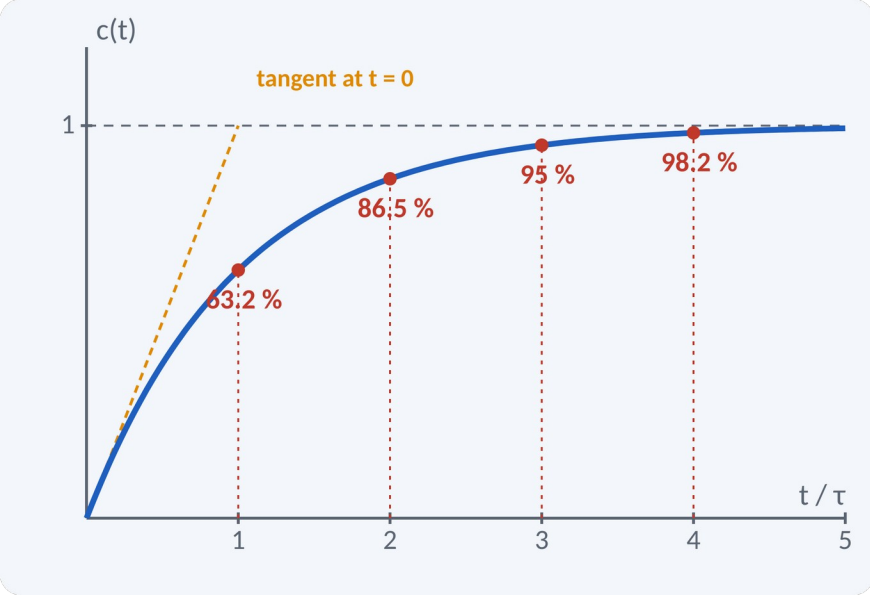
- Using 63.2 % of the input instead of the final output.
- Taking ts = 4τ for a 5 % band.
- Forgetting the DC gain K in c(∞).
- Mixing τ with the pole value 1/τ.
- Expecting overshoot in a first-order system.

Exam tips

- Convert to the form K/(τs + 1) first.
- Read K and τ, then use the formulas.
- Check units of τ (seconds).
- Remember ess(ramp) = τ for unity-gain first order.

Summary

- Test signals: step, ramp, parabola, impulse, sinusoid.
- Response = transient + steady state.
- Zero order: c(t) = K·r(t).
- First order: c(t) = 1 − e^(−t/τ); ts = 4τ (2 %).
- Ramp error = τ; impulse is derivative of step response.



G(s) = K/(τs + 1); c(t) = K(1 − e^(−t/τ))

Quick Quiz

Q1. Laplace transform of the unit ramp is:
(a) 1 (b) 1/s (c) $1/s^2$ (d) $1/s^3$

Ans: (c)

Q2. At t = τ a first-order step response reaches:
(a) 50 % (b) 63.2 % (c) 86.5 % (d) 98 %

Ans: (b)

Q3. ts (2 %) of a first-order system with τ = 3 s is:
(a) 3 s (b) 9 s (c) 12 s (d) 6 s

Ans: (c)

Q4. Steady-state error of 1/(τs+1) for a unit ramp is:
(a) 0 (b) τ (c) 1/τ (d) ∞

Ans: (b)

Q5. A zero-order system has:
(a) one pole (b) two poles (c) no poles (d) a pole at origin

Ans: (c)

References

1. **I. J. Nagrath and M. Gopal, Control Systems Engineering**
   5th Ed., New Age International (P) Ltd., 2008 (Units 1, 2 textbook)

2. **K. Ogata, Modern Control Engineering**
   5th Ed., Pearson Education Asia / PHI, 2009 (reference)

3. **P. S. Satyanarayana, Concepts of Control Systems**
   2nd Ed., Dynaram Publishers, Bangalore, 2001 (reference)

Next lecture: second-order response (done) and then static error constants